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RESEARCH

Underwater acoustic metalens for enhanced sound transmission penetration through an air layer

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Abstract

Supercavitation technology is an underwater drag reduction technique that generates a sustained gas envelope around an underwater vehicle to overcome viscous drag, thereby achieving approximately 90% drag reduction. However, severe impedance mismatch inhibits sound waves from penetrating the underwater air layer, resulting in acoustic transmission losses of over 60 dB which critically limits the communication capabilities of supercavitation underwater vehicles. Owing to the need to overcome the constraints imposed by the two water-air interfaces, nearly no acoustic devices can enhance acoustic transmission without disrupting the cavity. To address this challenge, we propose a design method for a laminated acoustic metalens based on the effective medium theory. This method enables efficient sound transmission with a certain operating bandwidth through the underwater air layer, which is primarily attributed to the control of phase differences. In this design, the laminated acoustic metalens independently achieves two key functionalities: transmission enhancement with a certain operating bandwidth through the underwater air layer and enabling tunability of the operating frequency. Compared with the bare underwater air layer, experimental results show that the laminated acoustic metalens achieves sound transmission enhancement of more than 30 dB within a 400 Hz operating bandwidth. Additionally, the proposed laminated metalens can realize transmission enhancement within the range where the incident angle is less than 80°,and the transmission frequency can be flexibly adjusted. Its exceptional performance demonstrates that the proposed metalens design methodology holds significant potential for navigation and guidance applications in underwater supercavitation vehicles.

Keywords Acoustic metalens · Acoustic metamaterial · Supercavitation technology · Effective medium · Sound penetration enhancement

1 Introduction

The hydrodynamic drag experienced by underwater vehicles is approximately 850 times greater than their aerodynamic drag in air. This substantial resistance fundamentally limits the maximum attainable velocity for conventional underwater propulsion systems [1]. Supercavitation technology represents a revolutionary approach to hydrodynamic drag reduction (achieving approximately 90% drag reduction). This technique operates by generating and maintaining a stable air-filled cavity surrounding the underwater vehicle surface, thereby overcoming viscous hydrodynamic resistance. The resultant drag reduction enables significant velocity enhancement, making supersonic underwater navigation theoretically achievable for underwater vehicles [2–4]. Torpedoes and projectiles developed using supercavitation technology can break through the limits of underwater movement, greatly increasing their range [5]. However, while the cavity reduces resistance, it also hinders the transmission of information. Due to the extremely rapid attenuation of underwater electromagnetic waves, acoustic communication is the most efficient and practical means of communication for underwater vehicles [6–8]. However,

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when sound waves hit the water-air interface, they lose about 99.9% of their energy due to the huge difference in impedance between air and water. For supercavitation vehicles, the presence of two water-air interfaces results in greater sound energy loss. This issue has long hindered the development of supercavitation vehicles and supercavitation torpedoes, to the extent that researchers have been unable to effectively control their trajectories. In summary, guidance and navigation of supercavitation vehicles have become one of the core issues in supercavitation technology research. The cavity is a double-edged sword for supercavitation vehicles, providing efficient drag reduction while hindering sound transmission. Therefore, breaking through this invisible barrier to construct a high-speed channel for sound transmission is a major challenge.

In recent years, the development of acoustic metamaterials [9–11] and acoustic metasurfaces [12–14] has provided new insights into solving this technical challenge. Acoustic metamaterials, as a novel type of artificial structure, are characterized by carefully designed microscopic or macroscopic units that enable them to exhibit special properties not found in natural materials. Examples include negative mass density [15, 16], negative bulk modulus [17, 18], and extreme Poisson's ratio [19, 20], which provide new degrees of freedom and enormous possibilities for controlling sound waves. Reasonably designed metamaterials can achieve many unique physical phenomena and functions, including acoustic focusing [21, 22], acoustic cloaking [23, 24], acoustic imaging [25, 26], sound insulation [27, 28], and sound absorption [29, 30]. Most existing acoustic metamaterials can only function in a single background medium, such as water or air. Due to the different physical properties of water and air, freely manipulating acoustic propagation in underwater environments remains challenging. Early research has shown that Helmholtz resonators can be used to create an underwater metamaterial with negative acoustic bulk modulus, laying the foundation for the realization of multiple functions underwater [31]. Based on underwater transform acoustics, researchers have proposed metamaterials such as pentamode metamaterials [32] and underwater acoustic carpets [33], successfully achieving underwater acoustic stealth. In the field of underwater acoustic beam focusing, an underwater biomimetic acoustic model was proposed based on effective medium theory and biomimetic acoustic conversion. By combining artificial metamaterials with animal biological sonar, directional emission and target detection were successfully achieved [34, 35]. In addition, low-frequency underwater acoustics have attracted considerable interest from researchers, who have designed underwater sound-absorbing metasurfaces such as local resonance underwater acoustic metamaterial absorbers [36], underwater acoustic metasurface absorbers [37], and impedance-matched underwater acoustic absorbers [38].

Due to the significant difference in impedance between water and air, only a few sounds energy can penetrate through the water-air interface. To enhance water-air sound transmission, many metamaterials units have been designed, including metasurfaces with load membranes [39], resonant bubbles immersed in water [40–42], topologically optimized rigid metasurfaces [43], metasurfaces with phase engineering [44], and impedance-matched metafluids [45]. However, these metamaterials are limited to crossing single-layer gas-liquid interface. Currently, there is no simple and feasible method to enable sound to break through the natural barrier of the underwater air layer, i.e., to achieve transmission enhancement across dual-layer gas-liquid interface.

In this work, we theoretically propose a design method for a laminated tunable acoustic metalens based on the effective medium principle, aiming to achieve efficient sound transmission with a certain operating bandwidth across underwater air layers. The designed metalens consists of two primary components: the upper layer is an air-based metalens sealed with thin films, whose precisely engineered structural parameters allow accurate tuning of transmission frequencies; the lower layer is a water-based metalens formed by the parallel coupling of two distinct units, where the introduction of synergistic coupling effects broadens the metalens' transmission bandwidth. These two components are vertically stacked in series along the thickness direction, thereby enabling highly efficient and tunable acoustic transmission with a certain operating bandwidth across the underwater air layer. This layered architecture exhibits exceptional design flexibility and customizability, effectively overcoming the limitations of a single metalens. Compared with the bare underwater air layer, experimental results demonstrate that the laminated acoustic metalens achieves sound transmission enhancement of up more than 30 dB within the operating bandwidth of 400 Hz (From 30.3 kHz to 30.7 kHz). The proposed design method realizes multi-frequency and efficient acoustic transmission across underwater air layers with a simple structure. Additionally, the proposed laminated metalens can achieve transmission enhancement within the range where the incident angle is less than 80°, and its transmission frequency can be flexibly adjusted. Furthermore, as this non-invasive design does not disrupt the underwater air layer, it holds significant potential application value in fields such as navigation and guidance for supercavitation underwater vehicles.

2 Theory model

Figure 1a presents a schematic diagram of enhanced sound transmission achieved via the laminated acoustic metalens. This metalens is attached to the surface of an underwater

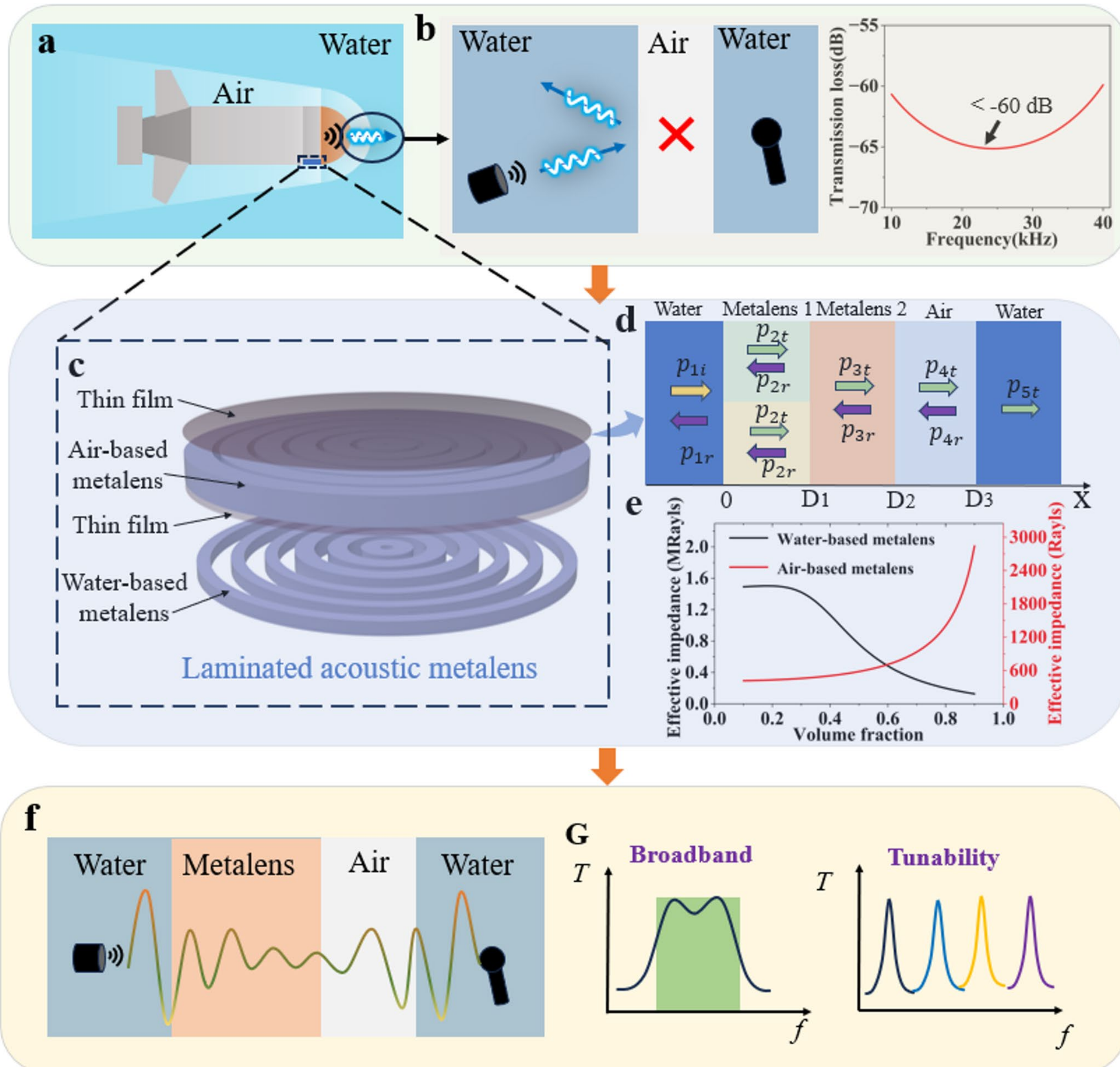

Fig. 1 The proposed concept of the laminated acoustic metalens for enhanced sound transmission across underwater air layers (**a**) Schematic of acoustic communication in the air layer under water; (**b**) Schematic diagram of sound transmission defects, showing that it cannot penetrate the underwater air layer, which has an extremely low sound transmission coefficient; (**c**) Schematic diagram of laminated acoustic metalens and conformal acoustic metalens; (**d**) The proposed multi-layer medium sound transmission model based on the transfer matrix method; (**e**) The calculated results of effective impedance for air-based metalens and water-based metalens; (**f**) Schematic diagram of tunable transmission of sound in underwater air layers through acoustically engineered metalens with non-invasive design

vehicle to enhance acoustic transmission across the air layer. As shown in Fig. 1b, nearly all sound incident on the underwater-air interface is reflected [46]. Owing to a significant impedance mismatch (approximately 3600 times), each water-air interface introduces an energy loss of about 30 dB. The cumulative effect of the two interfaces thus leads to a total attenuation of approximately 60 dB, resulting in a near-zero transmission coefficient and effectively preventing sound from propagating through the air layer. To verify the feasibility of the proposed metalens concept depicted in Fig. 1c, a multi-layered medium sound transmission model was established based on the effective medium theory. The propagation behavior of sound within the underwater air layer is visualized in Fig. 1d. The metalens 1 and 2 correspond to the water-based metalens and the air-based metalens of the laminated metalens in Fig. 1c, respectively.

The metalens is considered to be a uniform medium with characteristic impedance R_m and wave number k_m. Its sound field consists of transmitted wave p_{2t} and reflected wave p_{2r}. Specifically, it can be expressed as

$$p_{2t} = p_{2ta}e^{j(\omega\, t - k_2 x)} \tag{1}$$

$$p_{2r} = p_{2ra}e^{j(\omega\, t + k_2 x)} \tag{2}$$

The particle velocity in the layer 2 of medium is

$$v_{2t} = v_{2ta}e^{j(\omega\, t - k_2 x)}) \tag{3}$$

$$v_{2r} = v_{2ra}e^{j(\omega\, t + k_2 x)} \tag{4}$$

Similarly, the sound field and particle velocity in the other layers of medium can be derived. Applying the sound pressure continuity condition and normal particle velocity continuity condition at $x = D_1$, we obtain

$$p_{2t} + p_{2r} = p_{3r} + p_{3t} \tag{5}$$

$$v_{2t} + v_{2r} = v_{3r} + v_{3t} \tag{6}$$

Combining the above equations, we obtain

$$\begin{bmatrix} p_{2ta} \\ p_{2ra} \end{bmatrix} = \frac{1}{2}A_2 \begin{bmatrix} p_{3ta} \\ p_{3ra} \end{bmatrix} \tag{7}$$

$$A_2 = \begin{bmatrix} \left(1 + \frac{R_2}{R_3}\right)e^{jk_2d_2} & \left(1 + \frac{R_2}{R_3}\right)e^{jk_2d_2} \\ \left(1 - \frac{R_2}{R_3}\right)e^{jk_2d_2} & \left(1 + \frac{R_2}{R_3}\right)e^{jk_2d_2} \end{bmatrix} \tag{8}$$

where d_2 is the thickness of the layer 2 of medium and R_3 is the characteristic impedance of the layer 3 of medium. Therefore, according to the recursive relationship (7), it can be easily derived that

$$\begin{bmatrix} p_{1ta} \\ p_{1ra} \end{bmatrix} = \left(\frac{1}{2}\right)^3 A_1A_2A_3A_4 \begin{bmatrix} p_{5ta} \\ p_{5ra} \end{bmatrix} \tag{9}$$

According to the multi-layer medium acoustic transmission model shown in Fig. 1e, $p_{1ta} = p_{1ia}$ and $p_{4ra} = 0$ in Eq. (9). Then, it can be rewritten as

$$\begin{bmatrix} p_{1ia} \\ p_{1ra} \end{bmatrix} = \left(\frac{1}{2}\right)^3 A_1A_2A_3A_4 \begin{bmatrix} p_{5ta} \\ 0 \end{bmatrix} \tag{10}$$

As the number of medium layers increases, the analytical solution for the sound transmission coefficient becomes increasingly complex. However, the impedance characteristics of the metalens can be used to simplify the calculation. As shown in Fig. 1e, the impedance and sound velocity of the air-based metalens are close to those of air, while the impedance and sound velocity of the water-based metalens are close to those of water. To simplify the calculation process, we analyze the air-based metalens and water-based metalens separately, reducing the five-layer medium to a four-layer medium. Specifically, when analyzing the air-based metalens, the water-based metalens is treated as equivalent to water. When analyzing the water-based metalens, the air-based metalens is treated as air. Since medium 1 is the same as medium 5, i.e., $R_1 = R_5$, combining the above equations yields the sound transmission coefficient as

$$T_p = \frac{|p_{5ta}|}{|p_{1ia}|} = \frac{8}{\sqrt{X^2+Y^2}} \tag{11}$$

$$\begin{aligned} X = 8cosk_2d_2cosk_3d_3 - \\ 4\left(\frac{R_m}{R_a} + \frac{R_a}{R_m}\right)sink_2d_2sink_3d_3 \end{aligned} \tag{12}$$

$$\begin{aligned} Y = 4\left(\frac{R_m}{R_w} + \frac{R_w}{R_m}\right)sink_2d_2cosk_3d_3 + \\ 4\left(\frac{R_a}{R_w} - \frac{R_w}{R_a}\right)cosk_2d_2sink_3d_3 \end{aligned} \tag{13}$$

where subscripts *w*, *a*, and *m* represent water, air, and metalens, respectively. The detailed derivation is provided in Supplementary Material S1. The simulation results of the laminated metalens well demonstrate the rationality of this simplification, which will be discussed in detail in Sect. 3.1. The above formulas reveal the transmission laws of sound waves in multilayer media. The sound transmission coefficient is related to the impedance, thickness, and wave beam of each layer of the medium. Besides, with the help of the multi-layer medium sound transmission model, we can predict the transmission frequency of the metalens pretty well. The above formula is rather complex, it's difficult to directly find the relationship between the sound transmission coefficient T_p and the transmission frequency. The transmission frequency can be computationally predicted using MATLAB numerical analysis.

Furthermore, the water-based metalens are introduced to significantly broaden the operational frequency band. In this work, the two different water-based metalens units are arranged in parallel on the plane normal to the direction of sound wave propagation. Its principle is to utilize the synergistic coupling effect between different water-based metalens units. The synergistic coupling effect refers to the phenomenon where two water-based metalens units generate two transmission frequencies due to their phase difference, thus exerting a synergistic effect. However, given the similar impedances of the two water-based metalens units, a coupling effect occurs, resulting in the shift of transmission frequencies. Therefore, precise design of structural parameters is necessary to couple the two transmission frequencies at the same position, thereby broadening the operating bandwidth. Furthermore, the coupling effect becomes more significant as the number of units increases. As illustrated in Fig. 1d, when sound propagates through two different parallel-connected media, it can be treated as propagating along two independent paths. Owing to the different sound velocities in the two media, a phase difference will arise even if the sound propagation paths are identical. Consequently, the transmitted sound field corresponds to the vector superposition of the two propagation paths. Equation (11) demonstrates that the perfect transmission frequency varies with factors such as impedance and wave beam. Accordingly, two sound propagation paths will yield two perfect transmission frequencies. Naturally, the sound transmission coefficient can also be theoretically calculated, and the transmission frequency and bandwidth can be predicted. When sound passes through two parallel media, its sound transmission coefficient can be expressed as

$$T_p = \frac{|p_{5ta}|}{|p_{1ia}|} = \sqrt{T_{p1}^2 + T_{p2}^2 + 2T_{p1}T_{p2}cos\phi} \tag{14}$$

where T_{p1} and T_{p2} are the sound transmission coefficients of the two different media, respectively, and ϕ is the phase difference of sound transmission. The specific derivation process is shown in Supporting Material S1. Considering the coupling effect between the two structures, a phase difference of $(2n+1)\pi$ will cause sound attenuation. Therefore, it is necessary to reasonably control the phase difference between the two paths, leverage synergistic effects to enhance sound transmission, and suppress the occurrence of transmission troughs. Although acoustic waves undergo severe attenuation when propagating through an underwater air layer due to extreme impedance mismatch, the introduction of the metalens at the water-air interface enables efficient and frequency-tunable sound transmission with a certain operating bandwidth. This result indicates that the proposed metalens concept, which is designed for multi-frequency and efficient sound transmission across underwater air layers, is feasible in principle.

3 Result and discussions

3.1 Design of the laminated acoustic metalens

As illustrated in Fig. 1c, the laminated acoustic metalens is composed of an air-based metalens sealed with a thin film and a water-based metalens, which are stacked along the thickness direction. The structural dimensions of each layer of the metalens are determined via theoretical and numerical calculations. This layered design integrates functions including tunability of the operating frequency and efficient transmission with a certain operating bandwidth, and offers extremely high design flexibility and customizability, thus overcoming the limitations of a single metalens. To verify the sound transmission enhancement performance of the designed laminated acoustic metalens, the sound transmission coefficient and transmitted sound field were calculated using the finite element numerical method.

To achieve tunable sound transmission with a certain operating bandwidth, it is only necessary to conduct separate designs based on the aforementioned theory. First, the air-based metalens and the water-based metalens are studied separately. Specifically, one type of metalens is used to replace the laminated metalens mentioned above. Finally, based on the similarity in impedance, they are studied together in a coupled manner. To address this fundamental challenge of tunable transmission, we propose an air-based metalens sealed with a thin film, whose specific structure is presented in Fig. 2a. Here, the lattice constant b is set to 5 mm. The expression for the volume fraction is $f = \frac{a^2}{b^2}$. By adjusting the edge length a of the solid inclusion while keeping other parameters constant, the effective impedance and sound velocity of the metalens can be flexibly tuned to meet the required acoustic impedance specifications. According to Eq. (11), the sound transmission coefficient of the metalens is closely associated with its impedance and sound velocity. To achieve perfect transmission which means a sound transmission coefficient of 1, set the required transmission frequency to obtain the impedance and sound velocity of the metalens. Since the impedance and sound velocity of the metalens are directly related to its volume fraction, the structural dimension parameters of the metalens can be indirectly obtained through the transmission frequency. The calculation method for the effective impedance of the metalens is detailed in Supporting Material S2. The theoretical derivation in Sect. 2 reveals that the smaller the acoustic impedance of the structure, the wider the adjustable range of the perfect transmission frequency. However, the impedance of previously reported structures [45, 47] is generally strongly correlated with the base medium, implying that the acoustic impedance only varies slightly around the value of the base medium. This poses significant challenges for achieving acoustic impedance as low as that of air in underwater environments, and such impedance is also difficult to control. In contrast to conventional architectures, we utilize PET thin-plate encapsulation to stabilize the air-based metalens underwater, thereby achieving low acoustic impedance and expanding the adjustable range of the perfect transmission frequency. This acoustic architecture overcomes the limitations of existing designs regarding environmental dependency and improves the stability of the

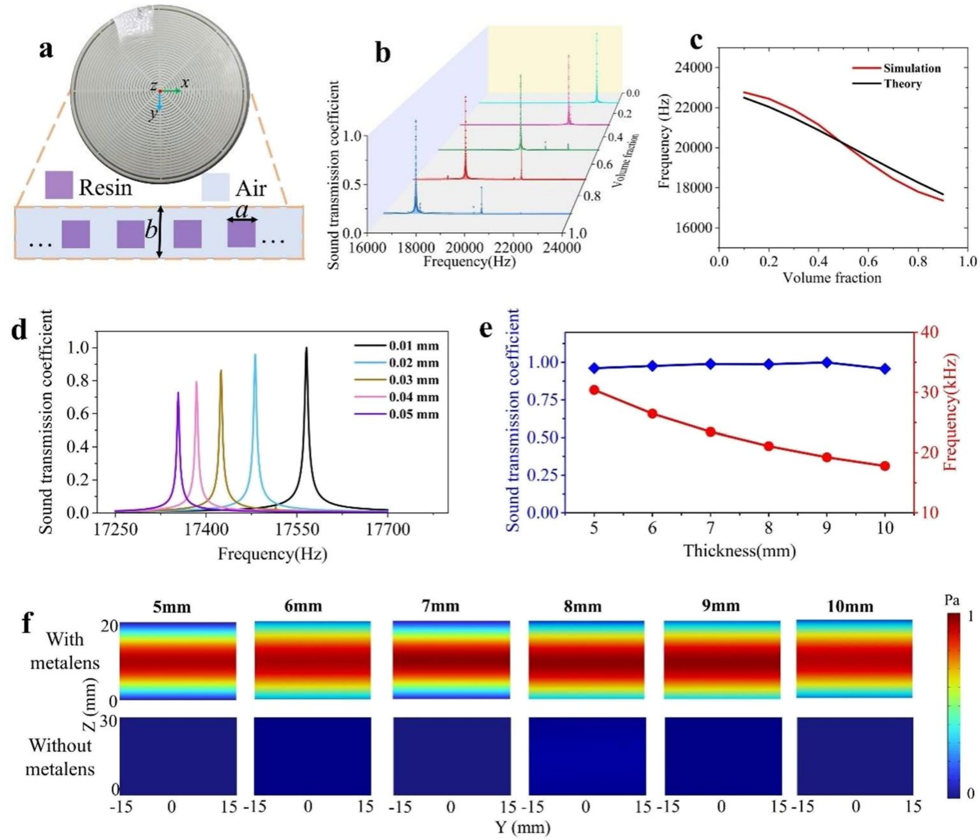

Fig. 2 The air-based metalens sealed with PET thin film (**a**) Schematic diagram of an air-based metalens, with a two-dimensional cross-section showing solid inclusions arranged periodically in the air; (**b**) The sound transmission coefficient calculation results for different volume fractions $f = a^2/b^2$ with the 10 mm thick air layer demonstrate the tunable characteristics; (**c**) Comparison of theoretical predictions and simulation results for perfect transmission frequency; (**d**) The thickness of PET thin films affects sound transmission, with thinner plates having less impact; (**e**) The sound transmission coefficient and perfect transmission frequency calculation results of air-based metalens under different air layer thickness conditions show that it have good transmission enhancement capabilities for air layers of different thicknesses; (**f**) Under conditions of the air layer thickness of 5–10 mm, the metalens enhances sound transmission relative to the bare underwater air layer

metalens. To better evaluate the transmission performance of the air-based metalens, numerical simulations were conducted on air-based metalens with different structural parameters. As illustrated in Fig. 2b, as the volume fraction of the air-based metalens decreases, the perfect transmission frequency shifts toward higher frequencies. It indicates that adjustment can be achieved over a wide operational range. Furthermore, the air-based metalens maintains a high sound transmission coefficient across different volume fractions, with the sound transmission coefficient approaching 1 when the volume fraction is 0.8. These results conclusively demonstrate the tunable characteristics of the designed metalens, which enables multi-frequency acoustic transmission. To validate the accuracy of the multi-layer medium sound transmission model, Fig. 2c presents the perfect transmission frequencies from analytical predictions (black line) and numerical simulations (red line), which are in good agreement. The analytical solution of the transmission frequency is obtained through calculations using Eqs. (11), (12), and (13). This indicates that the performance of the structure can first be predicted theoretically, facilitating the gradual optimization of the design. It also confirms that the multi-layer medium sound transmission model holds guiding significance for engineering applications.

Considering that PET thin films may affect sound transmission, we performed numerical simulations of air-based metalens with different thicknesses (PET thin films) in finite element software, as shown in Fig. 2d. It shows that PET thin films of different thicknesses have different transmission capabilities. The thinner the PET thin film, the stronger its transmission enhancement performance. When the thickness is 0.01 mm, the sound transmission coefficient is approximately 1, which is close to lossless transmission. In addition, the thickness of the PET thin film also affects the perfect transmission frequency, causing it to shift slightly toward lower frequencies. This is because the introduction of the PET thin film creates additional acoustic impedance, and the impedance mismatch increases acoustic energy loss. However, the calculation results in Fig. 2d show that sufficiently thin PET thin film cause almost no sound energy loss, and the effect of their acoustic impedance can be neglected. To verify the adaptability of air-based metalens, we set up air layers of different thicknesses in finite element software to simulate cavity generated by cavitators of different diameters. Here, we use an air-based metalens with a thickness of 5 mm and a volume fraction of 0.8 as an example for calculation. Figure 1e demonstrates that for air layers with 5–10 mm thickness, the air-based metalens consistently achieve sound transmission coefficients exceeding 0.95, allowing most acoustic waves to pass through the underwater air layer and achieve perfect transmission. The perfect transmission frequency decreases as the thickness of the air layer increases. To more intuitively illustrate the transmission enhancement performance of air-based metalens, Fig. 2f shows the normalized sound field distribution in the y-z plane above the metalens for different thicknesses of air layers. With the setup of the metalens, the transmitted sound pressure of all thicknesses of air layers showed a significant increase, and the incident sound waves effectively passed through the air into the water, further proving that air-based metalens can enhance sound transmission across underwater air layers. These results collectively indicate that thin-film sealed air-based metalens exhibit excellent transmission enhancement performance and are tunable when sound propagates through underwater air layers.

The importance of broadband transmission capability for acoustic communication is well-recognized. It can enhance communication efficiency and system capacity, and serves as the key to enabling high-speed underwater data transmission [48–50]. To achieve efficient sound transmission with a certain operating bandwidth across the underwater air layer, we next designed an impedance-controllable water-based metalens [47], whose structure is presented in Fig. 3a. The effective impedance and effective acoustic velocity of this water-based metalens were calculated using transmission coefficient inversion methods [47], and these parameters are also correlated with the metalens' volume fraction. The results of the effective impedance calculation for the water-based metalens are shown in Fig. 1e. The core concept of the broadband design method is to utilize synergistic coupling effects between different water-based metalens units to construct two water-based metalens units with appropriate dimensions, thereby broadening the metalens' transmission bandwidth. Here, the lattice constant b is also set to 5 mm, and the transmission capability of the metalens is adjusted by controlling the edge length a of the rubber solid inclusions and the number of units n. As illustrated in Fig. 3b, the transmission phase and sound transmission coefficient of water-based metalens units vary with their side lengths. The transmission phase can be tuned within the range of 0.8π to -0.6π, while the sound transmission coefficient remains above 0.5, with a maximum value of 0.8. On the one hand, the phase difference between the two units must be less than 0.5π to enhance the amplitude of the superimposed transmitted sound pressure. On the other hand, both units should exhibit high sound transmission coefficients, which constitutes the basis for broadening the frequency band. Therefore, Units A_1 and A_2 shown in Fig. 3b were selected as the basic units of the water-based metalens, with side lengths of 4 mm and 2.5 mm, respectively.

To verify the transmission enhancement performance of the water-based metalens, its sound transmission coefficient was calculated using finite element software. As shown in Fig. 3c, the sound transmission coefficient of the water-based metalens exhibits two distinct peaks. When multiple different types of metalens units are periodically arranged to form a metamaterial, significant acoustic coupling effects occur between adjacent units, i.e., the interaction and transmission of acoustic energy between units. Moreover, with the increase in the number of metalens units, the acoustic coupling effect between adjacent units is further enhanced. These peaks gradually converge as the number of units increases, thereby forming a certain bandwidth. However, when the number of units is large, the wavelength of sound waves in the metalens becomes shorter than the total length of one type of unit. According to the effective medium theory, sound waves cannot effectively distinguish between the two different media, thus inhibiting efficient transmission with a certain operating bandwidth. The theoretical data in the figure is obtained through calculations using Eqs. (11), (12), (13), and (14). It is evident that the two sound transmission coefficient curves exhibit the same trend with frequency, both displaying two transmission peaks which confirms the necessary consistency between theory and simulation. Minor differences exist between the two sets of results, however: the simulation results show a broader transmission frequency band, and the frequencies of the transmission peaks also differ. This is because our theoretical derivations assumed an infinite number of metalens units, whereas the actual number of metalens units is finite which affects the sound transmission coefficient of the metalens. The simulation results in Fig. 3c clearly confirms this point. To more intuitively demonstrate the transmission enhancement performance of the water-based metalens, Fig. 3e shows the normalized sound field distribution in the y-z plane above the metalens at different frequencies. In contrast to the air-based metalens, the acoustic pressure contour map above the water-based metalens reveals sound converging toward the metalens center, indicating a certain degree of acoustic focusing. This phenomenon arises because the water-based metalens comprises two distinct parallel-connected units, which generates a radial gradient refractive index and thus enables acoustic focusing. In summary, efficient sound transmission with a certain operating bandwidth across the underwater air layer can be achieved by carefully designing the size and number of solid inclusions in the water-based metalens.

The air-based metalens and water-based metalens were designed separately in the previous section; next, these two layers of metalens will be coupled to form a laminated metalens. As presented in Fig. 4a, by combining the air-based metalens and water-based metalens, we designed

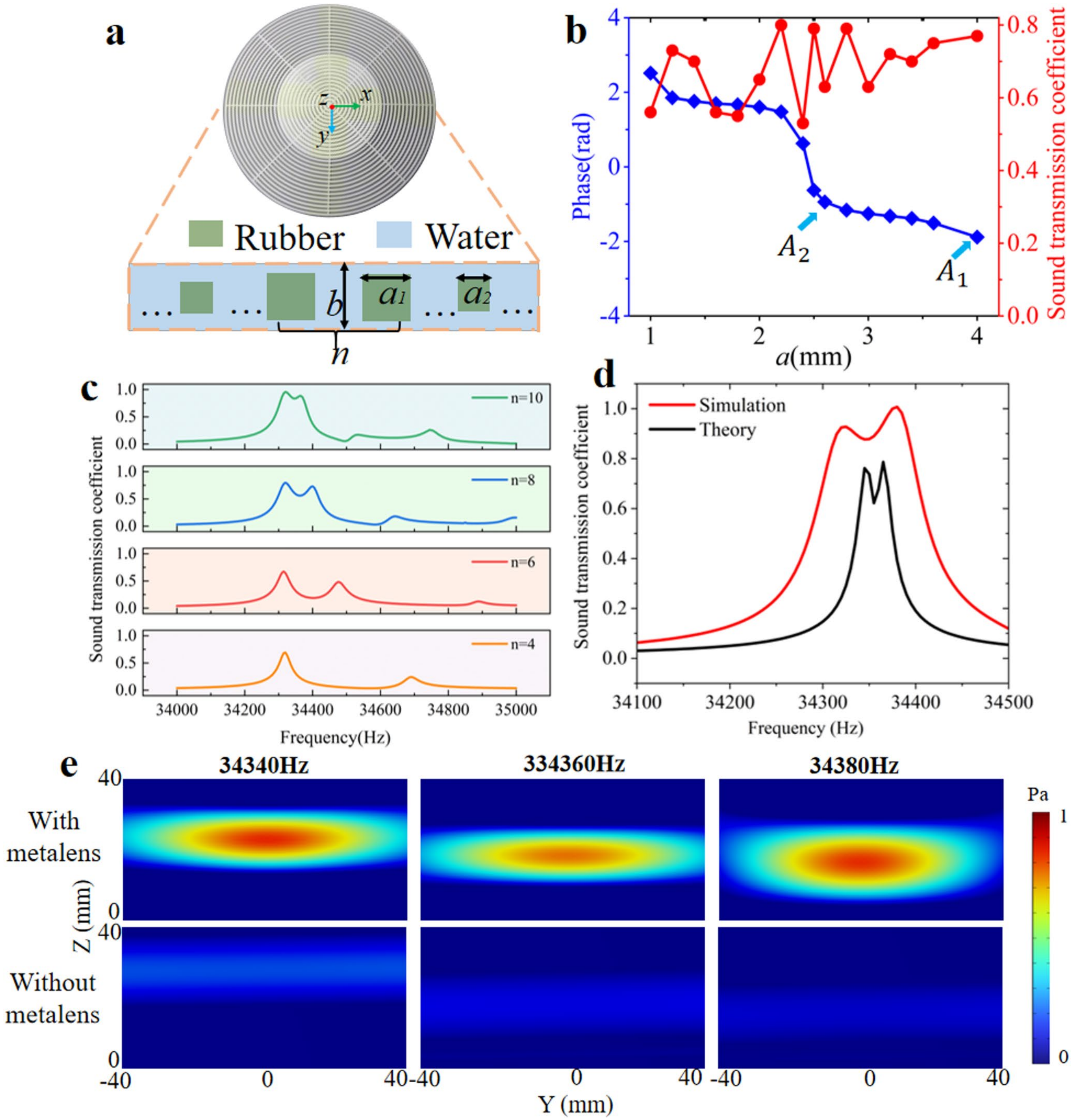

Fig. 3 The water-based acoustic metalens (**a**) Schematic diagram of water-based metalens, with two-dimensional cross-section showing solid inclusions arranged periodically in water; (**b**) The correlation between the phase and transmission coefficient of water-based metalens units and unit size, where $A_1$and A_2 represent the units used in the design; (**c**) The simulation calculation results of sound transmission coefficients for water-based metalens with different numbers of units; (**d**) Comparison of sound transmission coefficients between simulation (red line) and theory (black line); (**e**) At three representative frequencies, the normalized transmission sound field of the water-based acoustic metalens and the normalized transmission sound field without the metalens show the transmission enhancement of the metalens

a laminated acoustic metalens that enables efficient and simultaneously tunable sound transmission with a certain operating bandwidth across underwater air layers. Here, an integrated design is implemented, which involves stacking the two metalens in series along the thickness direction. To verify the efficient sound transmission enhancement performance of the laminated acoustic metalens, its sound transmission coefficient was calculated using finite element

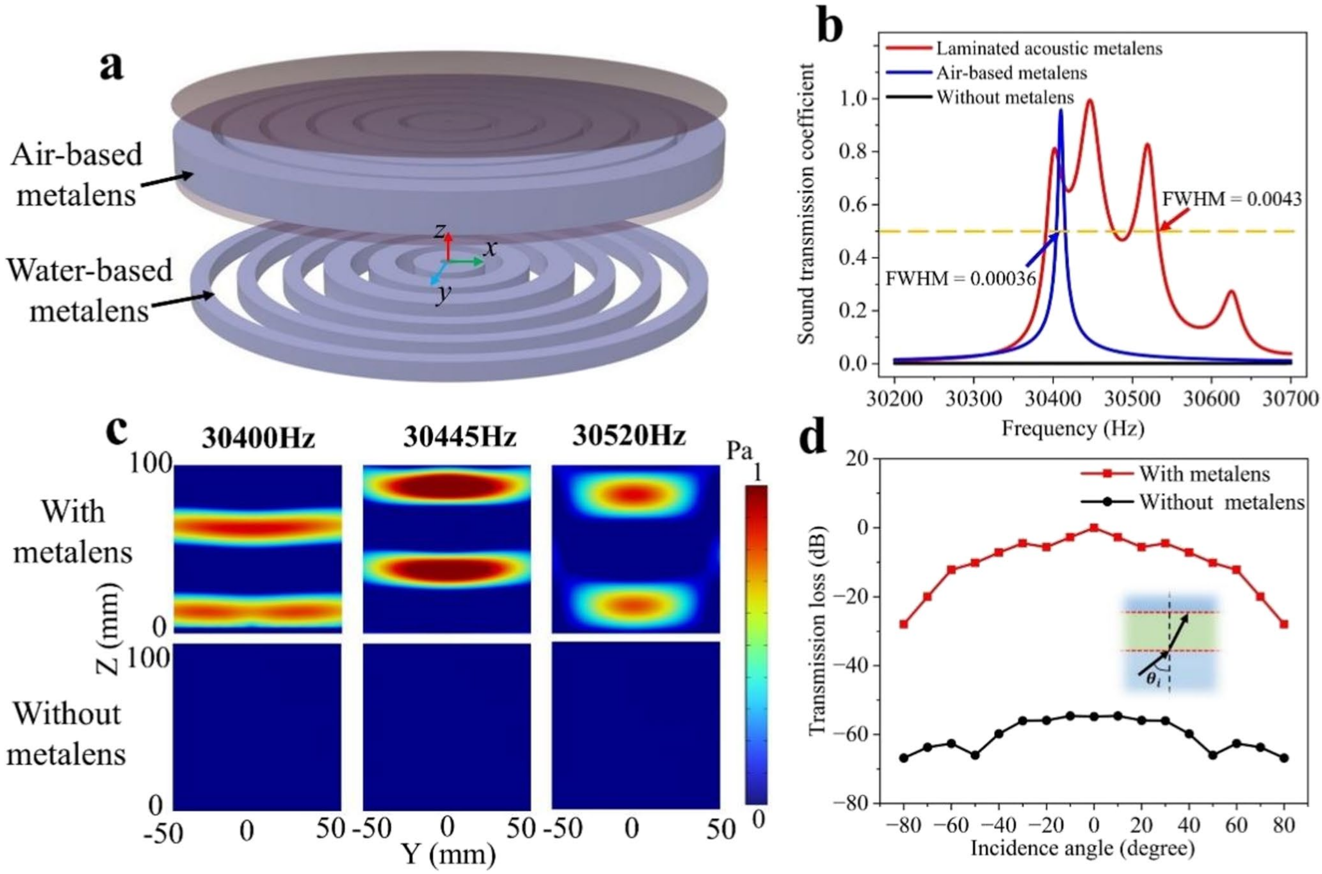

Fig. 4 The laminated acoustic metalens (**a**) Schematic diagram of the laminated metalens, composed of the air-based metalens and the water-based metalens stacked in the direction of thickness; (**b**) Comparison of sound transmission coefficients between laminated acoustic metalens (red line), air-based metalens (blue line), and no metalens (black line); (**c**) At three representative frequencies, the normalized transmission sound field of the laminated acoustic metalens and the normalized transmission sound field without the metalens show the transmission enhancement of the metalens; (**d**) The transmission losses for underwater air layers with and without the laminated acoustic metalens under oblique acoustic wave incidence

software. As shown in Fig. 4b, the sound transmission coefficient of the underwater air layer without the metalens is close to 0. When only the air-based metalens is used, the sound transmission coefficient exceeds 0.9, which greatly improves the sound transmission efficiency across the underwater air layer. However, the transmission bandwidth of the air-based metalens is extremely narrow, making it unsuitable for practical applications. Compared with single-layer structures containing only the air-based metalens, the laminated acoustic metalens broadens the transmission bandwidth, with the sound transmission coefficient exceeding 0.5 in the frequency range of 30,390–30,540 Hz. At a frequency of 30,445 Hz, the sound transmission coefficient is approximately 1, which is much higher than the bare underwater air layer. Since the absolute value of the bandwidth varies at different frequencies for the same method, the full width at half maximum (FWHM) remains nearly constant. The FWHM is calculated as the ratio of the peak width at half the peak height to the central transmission frequency. Therefore, the FWHM is employed to represent the broadband enhancement capability of the metalens. As shown in Fig. 3b, the FWHM of the laminated metalens is 12 times that of the air-based metalens, which demonstrates that the laminated metalens successfully broadens the operating bandwidth. Furthermore, the results presented in Fig. 4b indicate that the central transmission frequency of the laminated acoustic metalens is essentially consistent with that of the single-layer air-based metalens. This finding once again demonstrates that the transmission frequency can be independently adjusted by the air-based metalens. At the same time, it also demonstrates the rationality of coupling air-based metalens with water-based metalens.

To further demonstrate the efficient sound transmission enhancement performance with a certain operating bandwidth, the normalized simulated sound pressure field in the y-z plane above the metalens is presented at three representative frequencies, as shown in Fig. 4c. It can be clearly observed that, compared with the transmitted sound field without the metalens, the amplitude of the transmitted sound with the metalens is significantly enhanced, which primarily due to the combination of the air-based metalens and water-based metalens. Similarly, the laminated metalens can also achieve a certain degree of acoustic focusing. The above analysis indicates that the metalens exhibits excellent transmission efficiency with a certain operating bandwidth under normal incidence of sound waves. In practical scenarios, sound waves may propagate from any direction. Therefore, research on oblique incidence is crucial for advancing the practical application of multi-layer acoustic metalens. To assess the effect of the incident angle on the transmission performance of the laminated acoustic metalens, finite element software was used to simulate and calculate the acoustic transmission loss when the incident angle varied from −80° to 80°. As illustrated in Fig. 4d, when the absolute value of the incident angle is less than 70°, the acoustic transmission loss under oblique incidence is no higher than 20 dB. Even at an incident angle of 80°, where the transmission loss is approximately 30 dB, this performance is still significantly better than that of the underwater air layer without the metalens. The designed laminated acoustic metalens exhibit increased transmission loss as the incident angle increases, while maintaining stable sound transmission enhancement performance. This phenomenon occurs because the transmission loss of the bare underwater air layer also increases with the incident angle. This is mainly due to the extremely large impedance difference between water and air, which leads to a very small refraction angle. When sound waves are incident from water to air at different angles, they almost propagate vertically into the air. Consequently, the path length through the air layer changes very little, and the phase difference remains almost unchanged [41]. In summary, the laminated acoustic metalens achieves efficient sound transmission with a certain operating bandwidth across underwater air layers, and possesses wide-angle and tunable transmission capabilities.

3.2 Experimental verification

To verify the sound transmission enhancement capability of the laminated acoustic metalens for underwater air layers, experimental tests were conducted. The Fig. 5a shows the physical experimental setup and the 3D-printed experimental samples, with the water tank having dimensions of 1 m × 1 m × 1 m. The schematic diagram of the experimental setup is presented in Fig. 5b. Owing to the difficulty in effectively suppressing sound wave reflection with this setup, obtaining accurate sound transmission coefficients in the experiment is challenging. Therefore, to evaluate the acoustic transmission enhancement performance of the metalens, the transmission amplitude enhancement (*TAE*) at the underwater air layer is defined as

$$TAE = 20\log_{10}\frac{|p_t^1|}{|p_t^0|} \tag{15}$$

where p_t^1 and p_t^0 represent the average transmitted sound pressure with and without metalens near the underwater air layer, respectively. TAE is used to qualitatively demonstrate the sound transmission enhancement performance of the metalens, and the calculation results are shown in Fig. 5c. In the frequency range of 30,300–30,700 Hz, the experimentally measured transmission amplitude enhancement (*TAE*) exceeded 30 dB, with a maximum value of 55 dB, which is close to the 60 dB associated with perfect transmission. To more intuitively demonstrate the comparison results between the experiment and simulation, we compared the sound transmission coefficients calculated by the simulation with and without the metalens, and obtained the results of the enhanced transmission amplitude. Differences between the simulation and experimental results are evident, primarily in the TAE amplitude. These discrepancies arise from multiple factors. On the one hand, manufacturing errors are unavoidable; such errors lead to inconsistencies between the material parameters set in simulations and the actual material parameters, thereby influencing the experimental test results. For instance, the Young's modulus of rubber exerts a significant impact on the acoustic transmission enhancement performance of the metalens. On the other hand, reflections from the water tank walls may interfere with the test results, causing slight fluctuations in the measured transmission amplitude enhancement at specific frequencies. Nevertheless, both simulation and experimental results indicate that the transmission frequency bands are essentially consistent, and their trends with frequency variation are aligned. In summary, both sets of results confirm that the proposed laminated acoustic metalens exhibits excellent sound transmission enhancement with a certain operating bandwidth, enabling efficient sound transmission across the underwater air layer.

4 Discussion and Conclusion

When sound waves pass through the underwater air layer, the transmission loss is approximately 60 dB. This problem is the greatest obstacle to realizing acoustic communication for supercavitation underwater vehicles. By combining an air-based metalens sealed with the polyethylene terephthalate (PET) film and a water-based metalens, efficient sound transmission enhancement with a certain operating bandwidth

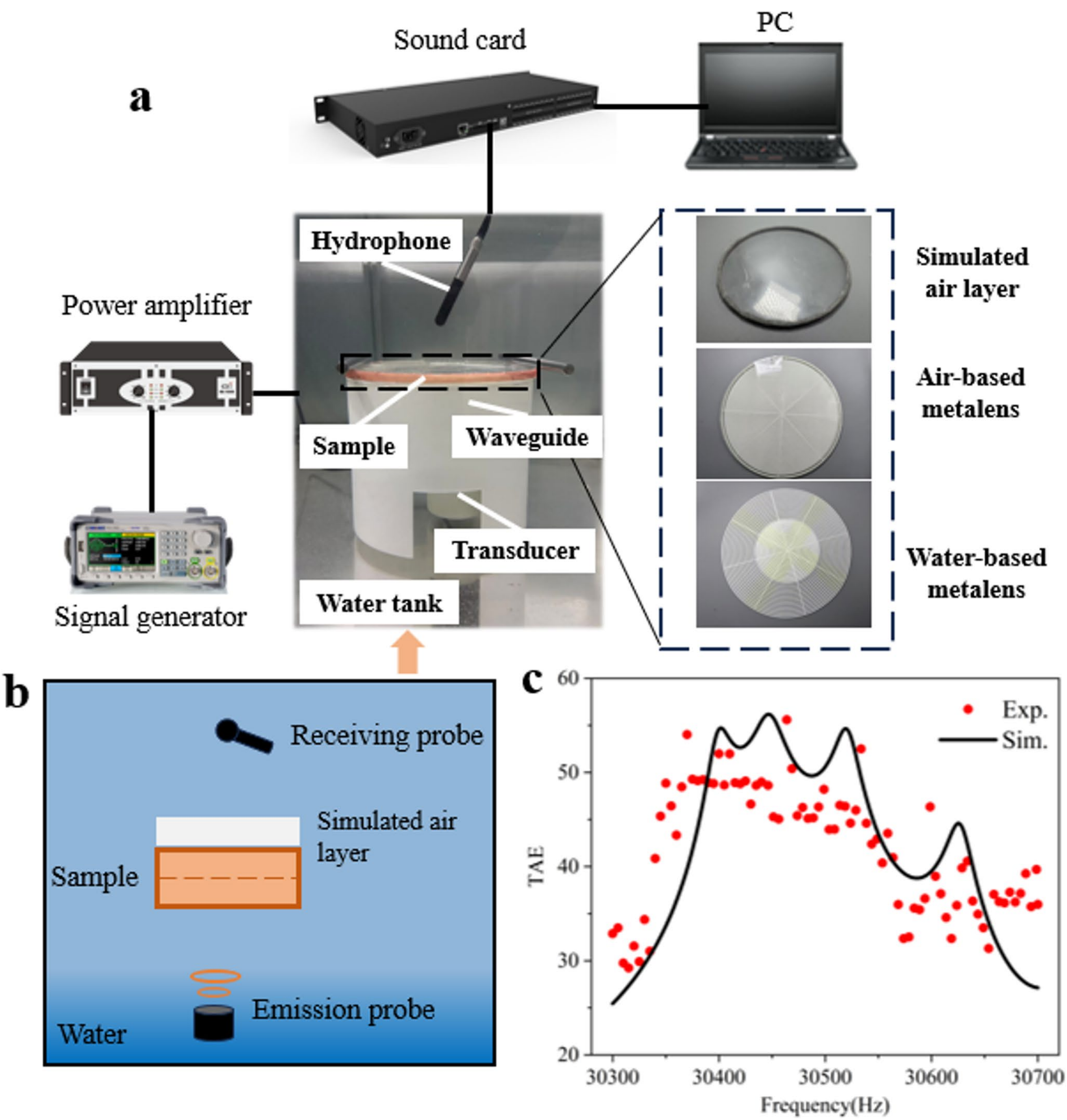

Fig. 5 Experimental verification of the laminated acoustic metalens (**a**) Schematic diagram of experimental equipment used to measure *TAE*, and the experimental equipment is shown only for schematic purposes. The inset on the right shows the 3D-printed experimental sample. (**b**) Schematic diagram of the experimental setup (**c**) The *TAE* experimental (Exp.) and simulation (Sim.) results for laminated acoustic metalens

across the underwater air layer is achieved. Precise design of its structural parameters enables accurate tuning of the metalens' transmission frequency, while the multi-layer medium sound transmission model allows for accurate prediction of this transmission frequency. Besides, the synergistic coupling effect between water-based metalens units broadens the operating bandwidth of the laminated metalens and the FWHM of the laminated metalens is 12 times that of the air-based metalens. Unlike existing research that focuses on single-layer water-air interfaces, this work is the first to investigate the transmission mechanism of sound propagating through underwater air layers.

The ideal application scenario of the metalens proposed in this study is to serve as a part of the cylindrical hull structure of the vehicle, directly replacing the traditional acoustic window structure. It will not interfere with the generation of the supercavity layer by the cavitator and ventilation holes at the head, and the generation and development process of the cavitation will not be damaged. Meanwhile, since the metalens itself undertakes the acoustic transmission function, the vehicle hull no longer requires additional acoustic windows or thick shielding in this region, so the acoustic signal will not be significantly attenuated due to the hull thickness. This study focuses on verifying the feasibility of the acoustic function of the proposed metalens, providing a new idea for the design of acoustic devices for supercavitating vehicles. In practical applications, multiple aspects such as relevant mechanical stability, pressure-resistant optimization design, and structural optimization design still need to be considered.

In conclusion, we propose and experimentally verify a design method for the laminated acoustic metalens. Both simulation and experimental results demonstrate that the laminated acoustic metalens successfully achieves efficient sound transmission with a certain operating bandwidth across underwater air layers. The proposed design method enables customized designs according to specific needs, such as for different operating frequencies or structural thicknesses. Meanwhile, the multi-functional integrated layered design also provides greater flexibility and customizability, primarily because each layer structure is implemented independently. It addresses the challenge of sound being unable to penetrate underwater air layers and provides a new approach for the study of acoustic communication in supercavitation underwater vehicles. The designed metalens features structural simplicity, high stability, and light weight, and holds important potential application value in fields such as navigation and guidance of supercavitation underwater vehicles. Additionally, the proposed multi-layer medium sound transmission model is applicable not only to water and air but also to other media—including air-solid, water-solid, and solid-solid systems. This opens up new possibilities for the design of novel cross-medium acoustic devices.

5 Method

5.1 Finite element numerical calculation method

The pressure acoustics and solid mechanics modules in COMSOL Multiphysics version 6.1 are used for numerical simulation. The laminated acoustic metalens uses a two-dimensional axisymmetric model to reduce computational costs. The background pressure field is set to 1 Pa to simulate the incident sound wave, and perfectly matched layers are added at the top and bottom to eliminate reflections. An acoustic-solid coupling boundary is applied at the interface between the metalens and the background medium, which couples fluid flow, structural stress analysis, and mesh deformation. The incident sound pressure and transmitted sound pressure are collected and calculated separately using the domain point probe method, and the two are divided to obtain the sound transmission coefficient. The specific parameters of the materials are as follows: air density and air sound velocity are ρ_a=1.21 kg/m^3, c_a=343 m/s, respectively; water density and water sound velocity are ρ_w=1000 kg/m^3, c_w=1500 m/s, respectively. The material of the air-based metalens solid inclusion is a photosensitive resin polymer with a mass density, Young's modulus, and Poisson's ratio of ρ_r=1320 kg/m^3, $E_r = 2.2\times 10^9 Pa$, and $\mu_r = 0.375$, respectively. The material of the water-based metalens solid inclusion is rubber, with a mass density, Young's modulus, and Poisson's ratio of ρ_r=1300 kg/m^3, $E_r = 2.0\times 10^6 Pa$, and $\mu_r = 0.49$, respectively.

5.2 Experimental setups

The air-based metalens was manufactured using photosensitive resin materials and light-curing 3D printing technology, and sealed using PET thin films. The air-based metalens was fabricated using photosensitive resin via light-curing 3D printing technology with subsequent encapsulation by PET thin film. The air-based metalens unit used in the laminated acoustic metalens has a side length of 4 mm, and the PET thin film has a thickness of 0.01 mm. The water-based metalens with a hardness of 50 HA was manufactured using a silicone rubber molding process. The two water-based metalens units have side lengths of 4.4 mm and 2.8 mm, respectively. Air-based metalens and water-based metalens are stacked in series in the thickness direction to form laminated acoustic metalens experimental samples. The air layer frame was manufactured using light-curing 3D printing technology and sealed with PET thin films to simulate an air layer underwater. Here, PET thin films have no effect on the transmission of sound in the underwater air layer and the detailed demonstration can be found in Supporting Material S3. In acoustic experiments, the experimental water tank measures 100 cm × 100 cm × 100 cm. As shown in Fig. 5a, the CT10-20 cylindrical transducer (Hangzhou Institute of Applied Acoustics, Hangzhou, China) located at the bottom of the water tank serves as the sound source. The air layer is positioned above the metalens and submerged underwater. The RHS10 hydrophone (Hangzhou Institute of Applied Acoustics, Hangzhou, China) is installed above the air layer to receive acoustic signals transmitted through the underwater air layer. The specific experimental procedure is as follows: the UTG962 signal generator [UNI-Trend Technology (China) CO., LTD of Songshan Lake, Dongguan City, Guangdong Province, China] is used to generate harmonic signals ranging from 10,000 to 40,000 Hz.

These signals are then amplified by UTC200 power amplifier (Hangzhou Institute of Applied Acoustics, Hangzhou, China) and converted into acoustic wave excitation by the transducer. Finally, the 16 A sound card (MOTU, American) inputs the acoustic signals collected by the hydrophone into a personal computer (PC) for data processing. The computer then performs Fourier transform on the collected time-domain signal to convert it into a frequency-domain signal. It then extracts the sound pressure amplitudes at different frequencies and calculates the transmission amplitude enhancement (*TAE*) through the underwater air layer.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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